

AI-Driven Computational Drug Discovery for Antimalarial Resistance: Integrating Chemical Space Exploration, Molecular Dynamics, and Machine Learning

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Abstract

Antimalarial drug resistance poses a critical threat to malaria control in Sub-Saharan Africa, with *Plasmodium falciparum* resistance emerging across multiple drug classes. This talk presents an integrated computational drug discovery pipeline combining artificial intelligence, molecular dynamics, and quantum-inspired methods to identify resistance-circumventing antimalarial candidates, with emphasis on African natural product chemical space.

Chemical Space Exploration and Virtual Screening. We established a chemically diverse library of 65,856 molecules through scaffold morphing and natural product hybridization, achieving 92.6% novelty relative to existing fingerprint space. Structure-based virtual screening against four validated antimalarial targets (*PfDHFR*, *PfCRT*, *PfATP4*, *PfClpP*) identified 20 top-ranked candidates (MPO 0.515–0.550). External validation through DEKOIS 2.0 benchmarking, MMV enrichment, and redocking (RMSD < 2.0 Å) confirmed protocol robustness.

Resistance-Aware Polypharmacology and Molecular Dynamics. We introduced a resistance retention score (RRS) framework for computational polypharmacology prioritization. Molecular dynamics simulations of 17 candidates across wild-type and mutant receptors (16 systems, 10 ns trajectories, OpenFF 2.2.0/CHARMM36m) revealed 87.5% static-versus-dynamic divergence, a positive estimand divergence serving as computational triage rather than failure. Deep learning scoring (GNINA CNN) achieved 100% Class-A agreement. African natural product profiling ($Fsp^3 = 0.22$, $QED = 0.70$, $MPO = 0.728$) provides regional context.

Quantum-Inspired Representations. Benchmarking topological fingerprints, network embeddings, and quantum kernels on 19,849 antimalarial-labeled compounds demonstrated equivalence rather than quantum advantage: ECFP4 achieved AUC 0.9475 ± 0.0045 versus 0.8876 (hybrid) and 0.8385 (quantum kernels), with no statistical superiority over classical RBF kernels ($p = 0.374$). This honest-negative result establishes rigorous standards: quantum-inspired descriptors provide complementary but not superior information.

Deep Learning Under Distribution Shift. Graph neural networks and transformers were benchmarked under scaffold-split validation. Classical ECFP4-random forest (AUC 0.8300 ± 0.0023) outperformed all learned representations: ChemBERTa (0.7867), GIN (0.8047), and topological fusion (0.8138). Nearest-neighbor decile stratification revealed fingerprint advantage concentrates in out-of-distribution regions (deciles 1–9), with convergence at high similarity (decile 10: ~ 0.88). External validation on 22,267 ChEMBL compounds confirmed this ordering. Topological fusion provides modest complementary signal but does not overcome generalization deficits at current scales, a second honest-negative result.

Methodological Contributions. Two key advances emerge: (1) *honest-negative reporting*—transparently documenting when advanced methods fail to outperform baselines provides essential guidance for method selection and computational resource allocation; (2) *estimand divergence framing*, computational disagreement (static docking versus molecular dynamics) serves as an informative filter rather than indicating failure. The pipeline provides computational hypotheses for experimental validation, emphasizing African natural products and WHO-documented Central African resistance profiles.

Future Directions. Experimental IC₅₀/EC₅₀ validation of top candidates, multi-target polypharmacology integration with advanced ML, and structure-phenotype fusion for mechanism-of-action prediction will advance translational impact for malaria-endemic regions.

Keywords: Antimalarial drug discovery, polypharmacology, resistance retention score, molecular dynamics, graph neural networks, quantum-inspired representations, honest-negative reporting, African natural products